



Modelling Financial Markets

Chapter 16: Digital Assets and DeFi



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BET: Bucharest Exchange Trading (index)
- BNB: BNB (native token of BNB Chain, formerly Binance Coin)
- BSC: BNB Smart Chain
- CAPM: Capital Asset Pricing Model
- CI: Confidence Interval
- CMKT: Crypto Market factor (value-weighted market excess return)
- CMOM: Crypto Momentum factor (past winners minus losers)
- COVID-19: COronaVirus Disease 2019
- CRIX: CRypto IndeX
- CSMB: Crypto Size factor (small minus big coins)
- DAI: Dai (stablecoin of the Maker/Sky protocol)
- DTB4WK: 4-Week Treasury Bill Secondary Market Rate (FRED series code)
- ECRIX: Exact CRIX (number of constituents chosen in steps of one)
- EFCRIX: Exact Full CRIX (global minimum of the AIC)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- ETHA: iShares Ethereum Trust ETF (ticker)
- FDIC: Federal Deposit Insurance Corporation
- FM: Fama–MacBeth (regressions)
- FRED: Federal Reserve Economic Data
- FWER: Family-Wise Error Rate
- GMM: Generalized Method of Moments
- GRS: Gibbons–Ross–Shanken (test)

Acronyms used in this material (2/2)

- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- IA: Internet Appendix (of the cited paper)
- LP: Liquidity Provider
- LUNA: Luna (native token of the Terra blockchain)
- LVR: Loss-Versus-Rebalancing
- MDD: Maximum DrawDown
- OLS: Ordinary Least Squares
- PC1: first Principal Component
- PCA: Principal Component Analysis
- SE: Standard Error
- SVB: Silicon Valley Bank
- TAR: Threshold AutoRegression
- TE: Tracking Error
- TMI: Total Market Index (all coins, CRIX methodology)
- TVL: Total Value Locked
- US: United States
- USDC: USD Coin (stablecoin issued by Circle)
- UST: TerraUSD (algorithmic stablecoin of the Terra blockchain)
- XRP: XRP (the native token of the XRP Ledger)

Chapter Route and Outcomes

Route of this chapter

- ▣ Part I: the crypto market, stylised facts of returns, crypto indices (CRIX), stablecoins
- ▣ Part II: DeFi and automated market makers, spot ETFs and crypto equities, cryptos as an asset class, a case study, one pass of the AI-assisted research loop
- ▣ Technical detail sits in the [Appendix](#) at the end

After this chapter you can

- ▣ measure a 7-day market on the right calendar and attach intervals to its tail index, volatility and correlations
- ▣ build a market-value index with a divisor and model a stablecoin peg as a threshold AR
- ▣ price a swap in a constant-product pool and explain why providing liquidity is selling volatility
- ▣ judge ETF tracking, crypto betas and asset-class claims with tests that respect closing times and unknown break dates

Blockchains, Coins and Tokens



Bust of Satoshi Nakamoto, Graphisoft Park,
Budapest

Photo: Fekist (2021); CC BY-SA 4.0; Wikimedia
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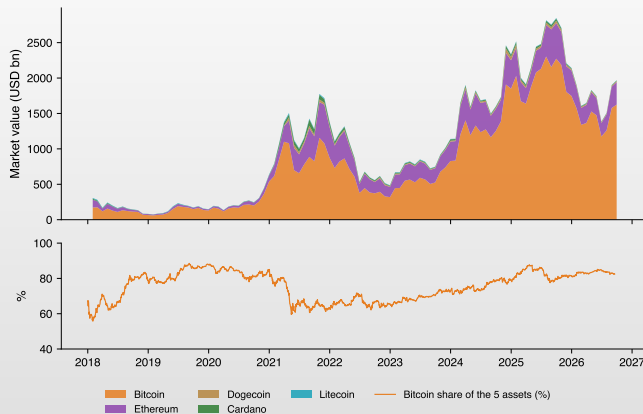
- **Blockchain Nakamoto (2008)**: a shared ledger of blocks, each carrying the hash of the previous one; secured by proof of work (Bitcoin) or proof of stake (Ethereum since 15 September 2022, ethereum.org: The Merge)
- Bitcoin issuance halves over time up to 21 million coins [Bitcoin.org](https://bitcoin.org) FAQ; 20.09 million existed on 18 Sep 2026
- **Coins** are native assets of a blockchain; **tokens** (stablecoins, governance tokens, tokenised funds) are issued by **smart contracts** Buterin (2014); DeFi (decentralised finance) is built from such contracts Schär (2021)
- Surveys: technology Lin, Pele & Ren (2025); economics Hårdle, Harvey & Reule (2020)

Data Used in This Chapter

Series	Content	Period
Crypto prices	Bitcoin, Ethereum, Solana, Dogecoin, Cardano, Litecoin; daily, 7 days a week	2018–2026
Stablecoins	USDT, USDC, DAI prices; supply by coin (DeFiLlama)	2019–2026
Coin supply	Coins in circulation, daily (Coin Metrics Community Data)	2018–2026
DeFi	Total value locked, all chains and by chain (DeFiLlama)	2020–2026
ETFs and shares	spot ETFs on Bitcoin (IBIT) and Ether (ETHA), Coinbase, Strategy; S&P 500, Nasdaq 100, gold	2018–2026
Asset universe	26 assets: cryptos, equity indices, currencies, commodities, bonds	2019–2026

- Daily closes up to 18 September 2026; crypto on its own 7-day calendar; analyses that combine assets use common days (prices aligned on common trading days first, then returns)

Five Coins, One Dominant Asset



- Market value $M_{i,t} = P_{i,t} \times Q_{i,t}$ (price in USD times coins in circulation), month ends; Bitcoin: 1.63 trillion USD on 18 Sep 2026
- The five coins: 1970 billion USD today, peak 3135 billion (6 Oct 2025); the value moves with prices, supply changes slowly
- Bitcoin's share: 82.5% today, between 55.8% (1 Feb 2018) and 88.5% (5 Sep 2019)
- Herfindahl–Hirschman index** $HHI = \sum_i w_i^2$ (w_i : value shares) = 0.71: an effective number of coins $1/HHI = 1.42$
- XRP and Chainlink are left out: their reported supply includes issuer-held tokens (free float matters)

Measures and the Tail Index

- Daily log return in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$, each series on its own calendar; $\sigma_a = s\sqrt{m}$, m = observations per year (about 365 for crypto, 252 for shares)
- Skewness S ; MDD = $\min_t (P_t / \max_{u \leq t} P_u - 1)$; volatility clustering $\hat{\rho}_1(|r|)$ (Chapter 1); historical daily $\text{VaR}_{1\%} = -q_{1\%}(r)$ and $\text{ES}_{2.5\%} = -\overline{r \mid r \leq q_{2.5\%}(r)}$ (Chapter 7)
- Hill estimator Hill (1975) from the k largest losses $L_{(1)} \geq \dots \geq L_{(k+1)}$, $k = 5\%$ of the losses:

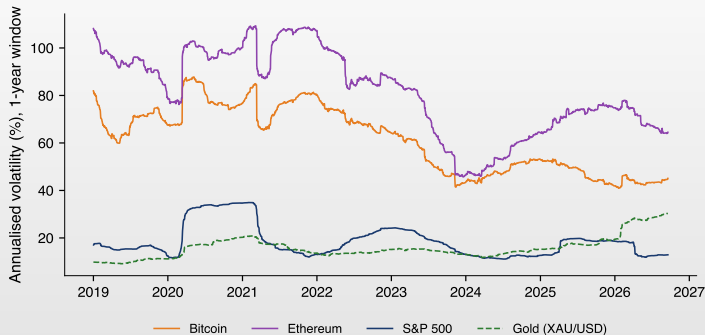
$$\hat{\alpha}_H = \left(\frac{1}{k} \sum_{j=1}^k \ln(L_{(j)} / L_{(k+1)}) \right)^{-1}$$
 - ▶ $P(L > x) \sim Cx^{-\alpha}$: smaller α = heavier tail; $\text{SE} = \hat{\alpha}_H / \sqrt{k}$ for independent and identically distributed (i.i.d.) data (conditions and dependence in the [Appendix](#))
- All six i.i.d. 95% CIs lie below 3.7: $\alpha < 4$, hence $E|r|^4 = \infty$, is plausible but not proven
 - ▶ If $\alpha < 4$, the sample excess kurtosis $K = \overline{(r - \bar{r})^4} / s^4 - 3$ estimates nothing: report $\hat{\alpha}_H$ with its CI instead
- Sample 2018–2026 (Solana from April 2020); comparison assets: S&P 500 and gold

Crypto vs Shares and Gold, 2018–2026

	n	σ_a (%)	S	MDD (%)	$\hat{\rho}_1(r)$	$\hat{\alpha}_H$	95% CI	VaR _{1%}	ES _{2.5%}
Bitcoin	3 182	64	−0.95	−82	0.17	2.73	[2.12, 3.33]	10.2	10.3
Ethereum	3 182	84	−0.82	−94	0.16	2.67	[2.08, 3.26]	13.5	13.7
Solana	2 351	117	−0.22	−96	0.28	2.85	[2.12, 3.58]	14.5	17.1
Dogecoin	3 182	124	4.88	−92	0.37	2.83	[2.22, 3.45]	13.8	16.4
S&P 500	2 189	19	−0.65	−34	0.36	2.87	[2.07, 3.67]	3.4	3.8
Gold	2 269	17	−0.55	−28	0.13	2.75	[1.99, 3.51]	3.0	3.2

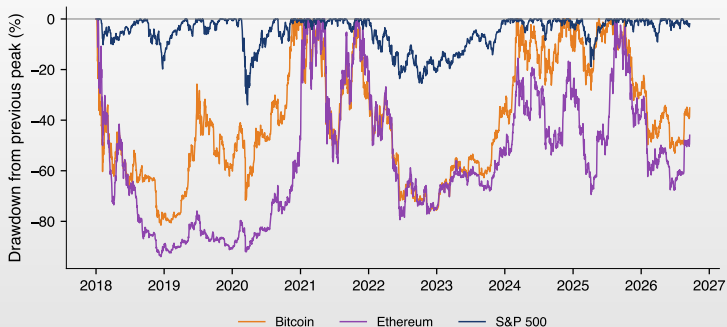
- Daily log returns in %; VaR and ES are losses of the daily log return; $\hat{\alpha}_H$ from the largest 5% of losses, CI $\hat{\alpha}_H \pm 1.96 \hat{\alpha}_H / \sqrt{k}$
- Bitcoin: 3.3 times the S&P 500's volatility; daily VaR 1% 10.2% in log terms, a loss of 9.7% of the position; worst day 12 Mar 2020 (COVID-19 crash)
- Drawdowns of 82–96% for the coins, 34% for the S&P 500; Hill indices 2.67–2.87 for all six assets: crypto is more volatile, but the data give no strong evidence of different tail indices (tests in the [Appendix](#))

Volatility Over Time



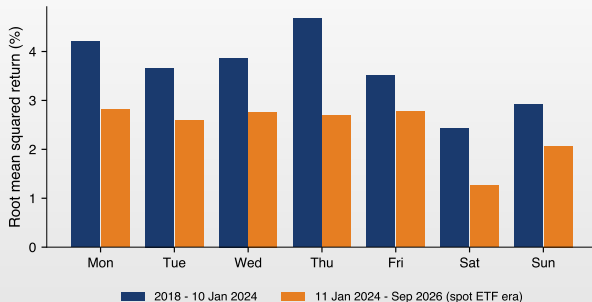
- Annualised standard deviation of daily log returns over a moving one-year window
- Bitcoin fell from 88% to 45% today, Ethereum from 109% to 64%: consistent with a maturing market, but the chart does not identify the cause
- Bitcoin is still 3.5 times as volatile as the S&P 500 (13%); gold reached 30% in 2026, a record for the sample

Drawdowns: How Deep and How Long



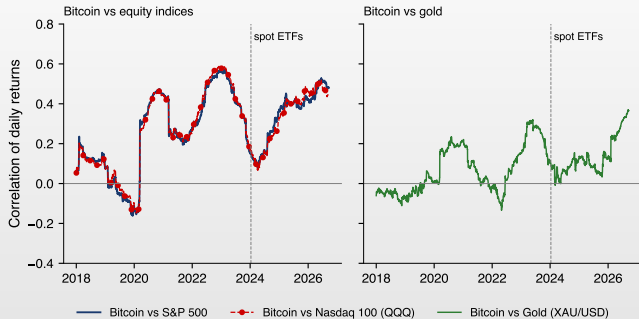
- Fall from the previous maximum, in %: Bitcoin was more than 50% below its peak on 36% of the days since 2018, Ethereum on 61%; the S&P 500 at most 34% (March 2020)
- Today: Bitcoin 35%, Ethereum 46% below their peaks; long “crypto winters” (2018–2020, 2022–2023) follow each boom (bubbles: [Chapter 17](#))

A 7-Day Market with a Weekend Lull



- Root mean squared daily Bitcoin return by day of the week; weekend variance / weekday variance: 0.45 before and 0.39 after 11 January 2024
- Bitcoin trades on Saturdays, but macro news, US equity trading and ETF flows do not; all days calmer after the ETF launch (weekday root mean square 4.01% → 2.73%)
- Annualise with 365 days, but do not treat all days as identical; did the ratio change? Seminar 16, B1

Is Bitcoin Moving with Shares?



- Correlation of daily log returns over 250 common days (prices aligned on common trading days first, then returns); Bitcoin vs S&P 500: 0.02 (2017–2019), 0.39 (2020–2023), 0.38 since the ETFs (674 days)
- Nasdaq 100 like the S&P 500 (0.38); gold -0.02 , 0.12 , 0.22 : a smaller diversification benefit than before 2020 [Bouri et al. \(2017\)](#), [Liu & Tsyvinski \(2021\)](#)
- The chart suggests a break in early 2020, not in January 2024: tested on the next two slides

Did the Bitcoin–Equity Link Break in 2020? A Formal Test

- Model on 2 692 common days since January 2016: $r_t^{\text{BTC}} = a + b r_t^{\text{SPX}} + u_t$, with (a, b) allowed to shift at an unknown date τ
 - ▶ sup-Wald [Andrews \(1993\)](#): $\sup_{\tau \in [0.15T, 0.85T]} W(\tau)$, $W(\tau)$ the Wald statistic for no change, with Newey–West covariances (Bartlett, 8 lags) in each regime
 - ▶ Under H_0 the sup is not χ_2^2 : its limit is $\sup_r \|B(r) - rB(1)\|^2 / [r(1-r)]$ (B a Brownian motion), simulated here: 5% critical value 11.55
- Result: $\sup W = 20.21$ ($p = 0.0008$); least-squares break date 9 Mar 2020 ($W = 18.81$); at the ETF launch $W = 2.31$
 - ▶ 95% CI for the date [Bai \(1997\)](#): 27 Nov 2018 – 15 Jun 2021; it excludes January 2024
 - ▶ Beta to the S&P 500: 0.07 before, 1.22 after; correlation 0.01 and 0.40
- Several breaks at unknown dates: the sequential tests and CIs of [Bai & Perron \(1998\)](#)

Correlation or Volatility? The Forbes–Rigobon Adjustment

- Pearson correlation rises mechanically when the volatility of the driving market rises, with the same underlying link [Forbes & Rigobon \(2002\)](#)
 - ▶ Adjusted correlation: $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, $\delta = \sigma_{\text{SPX,high}}^2 / \sigma_{\text{SPX,low}}^2 - 1$, with 2017–2019 as the calm base period
- 2020–2023: S&P 500 volatility 23.1% vs 12.8% ($\delta = 2.24$); $\hat{\rho} = 0.39$ becomes $\rho^* = 0.23$
 - ▶ Since the ETFs: $\hat{\rho} = 0.38$, $\rho^* = 0.33$ ($\delta = 0.42$)
- Change against 2017–2019 ($\hat{\rho} = 0.02$): $\rho_{2020-23}^* - \hat{\rho}_{2017-19} = 0.21$, 95% block-bootstrap CI [0.09, 0.34]
 - ▶ Under the adjustment's assumptions, 44% of the raw rise is attributed to volatility; the adjusted increase remains positive: stronger co-movement, not an identified mechanism
- Caveat: the adjustment assumes no endogeneity and no omitted common shocks, which a pandemic hardly satisfies

A Crypto Index and Its Divisor



Humboldt-Universität zu Berlin, where CRIX was developed

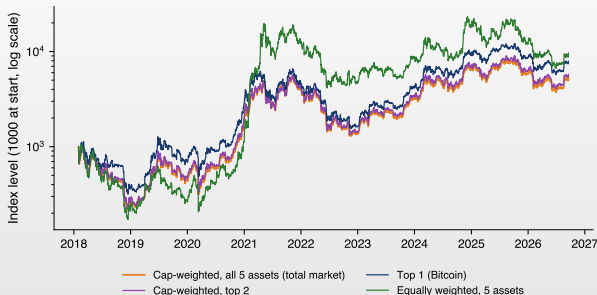
Photo: Diego Delso (2016); CC BY-SA 4.0;
Wikimedia Commons

- ▣ **CRIX Trimborn & Härdle (2018)**: a market-value weighted benchmark of the crypto market, published at thecrix.de; review of crypto indices: Găman, Pele, Härdle & Będowska-Sójka (2026)
- ▣ Level on day t of reallocation period τ : $I_t = \sum_{i=1}^k P_{i,t} Q_{i,\tau} / D_\tau$
 - ▶ $Q_{i,\tau}$: coins in circulation, fixed within the period; D_τ : the **divisor**
 - ▶ Index return = weighted average of the constituents' returns, weights = value shares at the start of the period; a Laspeyres-type index (quantities fixed within the period)
- ▣ At a reallocation (new Q , new members) the divisor keeps the level: $D_{\tau+1} = \sum_i P_{i,t} Q_{i,\tau+1} / I_t$
 - ▶ Here: monthly reallocation, weights from the previous month end; worked examples: Seminar 16, A1–A2

How Many Constituents? The CRIX Rule

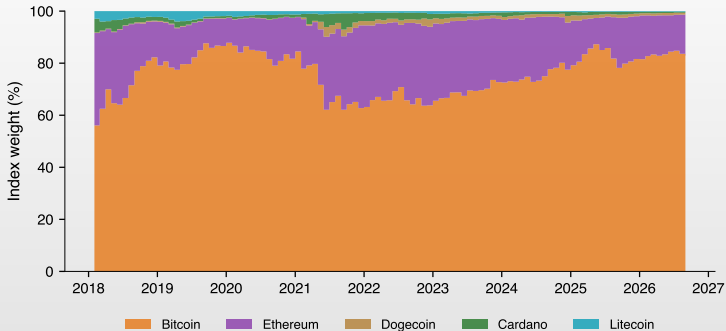
- Candidate index $\text{CRIX}(k, \beta)$: the k_1 largest coins at weight 1, the next $s = k - k_1$ with free weights β , fitted to track the total market index TMI (all coins) [Trimborn & Härdle \(2018\)](#)
 - ▶ Tracking residual $\hat{\varepsilon}_t(k, \beta) = \Delta \ln \text{TMI}_t - \Delta \ln \text{CRIX}_t(k, \beta)$
- $\text{AIC}(k) = -2 \log L\{\hat{\varepsilon}(k, \hat{\beta})\} + 2s$: more coins track better but cost parameters
 - ▶ k reviewed quarterly on the last three months of data; members reallocated monthly by market value
 - ▶ Two stopping rules, ECRIX (first rise of the AIC) and EFCRIX (global minimum); likelihood details in the [Appendix](#)
- Data limitation: circulating supply is public for five of our coins only (Bitcoin, Ethereum, Dogecoin, Cardano, Litecoin), so $k \leq 4$ and TMI has five members, not hundreds

Crypto Indices Since 2018



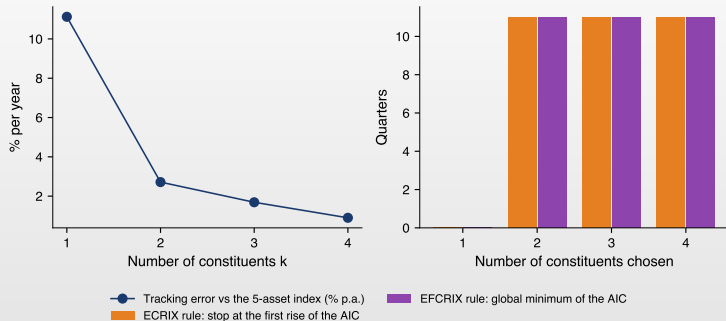
- Log scale; total-market index 21.4% a year over 8.6 years $((I_T/I_0)^{1/\text{years}} - 1)$, volatility 65.7%; Bitcoin alone did better: the other coins lost weight and value
- Equal weights: 30.0% a year at 82.5% volatility, driven by 2021 (+998% vs +113%): a few extreme months, not a stable premium
- An index is only as good as its universe: today's survivors are not the leaders of 2018 Będowska-Sójka, Wójcik & Pele (2026)

Index Weights Over Time



- Value weights at each monthly reallocation, %: Bitcoin 75.7% on average, never below 56.1%, today 82.9%; Ethereum 15.8%; the other three together below 2%
- Altcoins (coins other than Bitcoin) gain weight late in booms (2021) and lose it in busts: a value-weighted crypto index is mostly Bitcoin

Tracking Error and the CRIX Rule



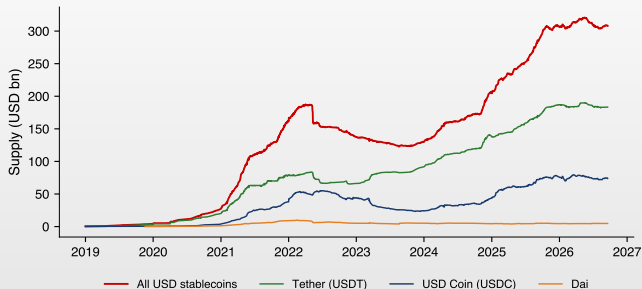
- Left: tracking error of the top- k index (weights 1) against the five-coin index: 11.1% a year for Bitcoin alone, 2.7% for two coins, 1.7% for three, 0.9% for four
- Right: the rule picks 2 coins in 11, 3 in 11 and 4 in 11 of 33 quarters (2 in the last); ECRIX and EFCRIX agree in every quarter

How a Stablecoin Holds 1 USD

Design	Backing	Examples (USD bn, 18 Sep 2026)
Fiat-backed	Bank deposits and short US Treasury bills held by the issuer	Tether (USDT) 183.33, USD Coin (USDC) 73.92, PayPal USD (PYUSD) 2.81
Crypto-backed	Over-collateralised crypto deposits or hedged positions, managed by smart contracts	Sky Dollar (USDS) 6.49, Ethena USDe 4.74, Dai (DAI) 4.77
Algorithmic	No external collateral: supply expands and contracts against a second token	TerraUSD (collapsed in May 2022)

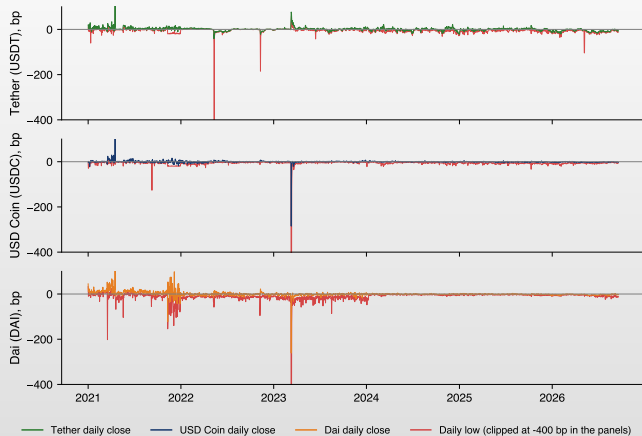
- Authorised traders **mint** (pay 1 USD, receive 1 coin) and **redeem** (return 1 coin, receive 1 USD) with the issuer: with costs c_r and c_m , arbitrage keeps the price in the band $1 - c_r \leq P \leq 1 + c_m$
 - Below the band: buy and redeem; above: mint and sell (evidence: [Lyons & Viswanath-Natraj \(2023\)](#); Tether issuance and Bitcoin prices: [Griffin & Shams \(2020\)](#))
- The band breaks when redemption stops or the backing is doubted; reserve and redemption rules: [MiCA, Regulation \(EU\) 2023/1114](#), [GENIUS Act](#), [Public Law 119-27](#)

The Rise of Stablecoins



- Supply in circulation ([DefiLlama](#)): from 4.2 billion USD in January 2020 to 308 billion on 18 Sep 2026, about 74 times: the cash leg of crypto trading and DeFi
- Tether 183 billion (60%), USD Coin 74 billion (24%); fiat-backed 283, crypto-backed 22.4, algorithmic 0.15 billion
- 2022–2023 dip: 187 billion (2 Apr 2022) to 123 billion (19 Aug 2023); TerraUSD vanished, USD Coin fell from 55 to 23 billion

Deviations from 1 USD



- Daily close and daily low minus 1 USD, in basis points ($1 \text{ bp} = 0.01\%$), 2021–2026; lows below -400 bp are cut at the panel edge
- Normal times: standard deviation of the close 7.3 bp (Tether), 7.5 (USD Coin), 10.8 (Dai); beyond 50 bp on less than 1% of the days
- Stress is rare but deep: USD Coin's low -1226 bp (11 Mar 2023), Tether's -515 bp (12 May 2022)
- A stablecoin is a claim with a short put on the reserves: nothing happens most days, a large loss is possible on a few

Arbitrage Bands Imply a Threshold AR

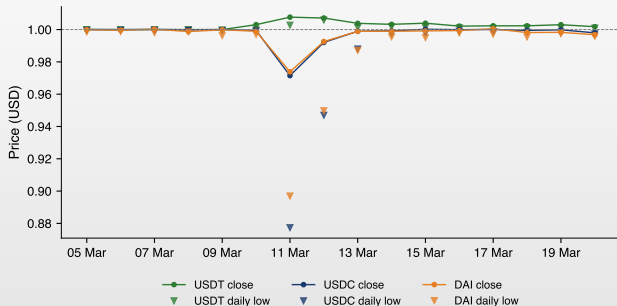
- Deviation from the peg in basis points (1 bp = 0.01%): $d_t = 10^4(P_t - 1)$, daily close
 - ▶ Inside the band $|d| \leq c$ arbitrage does not pay: weak pull; outside it does: fast reversion
- Threshold autoregression (TAR), the equilibrium form of [Balke & Fomby \(1997\)](#): $d_t = \phi_{\text{in}} d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$, $d_t = \phi_{\text{out}} d_{t-1} + \varepsilon_t$ otherwise
 - ▶ $\hat{c}$ by least squares over a grid of $|d_{t-1}|$ values, keeping at least 15% of the days in each regime
 - ▶ $\hat{c}$ (in bp) is an empirical regime boundary: it matches the costs only under a symmetric band $c_r = c_m$ (USD per token), as $10^4 c_r$ bp, and other frictions also move it
- Test of linearity $H_0: \phi_{\text{in}} = \phi_{\text{out}}$: c is not identified under H_0 , so $\sup_c W(c)$ is not χ_1^2 (the Davies problem)
 - ▶ p -value from the fixed-regressor bootstrap of [Hansen \(1996\)](#) with heteroskedasticity: $d_t^* = \hat{\varepsilon}_t \eta_t$, $\eta_t \sim N(0, 1)$, regressors held fixed; $W(c)$ with White standard errors
- Sample 2021–2026: earlier quotes are thin and noisy

Threshold AR Estimates for the Three Stablecoins

	$\hat{c}$ (bp)	$\hat{\phi}_{\text{in}}$ (SE)	$\hat{\phi}_{\text{out}}$ (SE)	sup W	p bootstrap	p (χ_1^2)	$\hat{\phi}$ linear
Tether	8.6	0.86 (0.02)	0.69 (0.11)	7.4	0.084	0.006	0.73
USD Coin	2.9	0.59 (0.04)	0.28 (0.03)	32.5	0.001	< 0.001	0.29
Dai	6.0	0.72 (0.08)	0.32 (0.07)	13.8	0.005	< 0.001	0.33

- USD Coin and Dai: linearity rejected; reversion is much faster outside the band (half-life 0.5 vs 1.3 days for USD Coin)
 - ▶ Without 9–16 March 2023, USD Coin gives $p = 0.29$ (bootstrap) but 0.019 (χ_1^2): one stress week carries the evidence, and χ^2 over-rejects
- Tether: the widest band ($\hat{c} = 8.6$ bp) and slow reversion inside it, but linearity is not rejected at 5%: consistent with fewer direct redeemers [Ma, Zeng & Zhang \(2025\)](#)

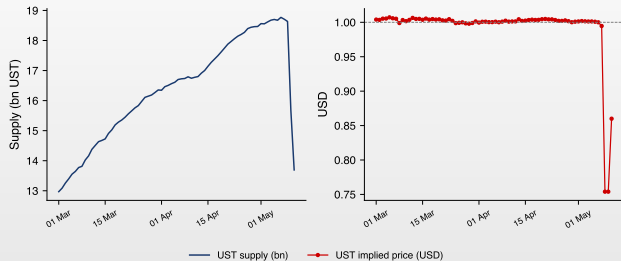
March 2023: a Bank Failure Hits USD Coin



- Daily close (lines) and daily low (triangles), 5–20 March 2023
- Friday 10 March: Silicon Valley Bank (SVB) is closed, the FDIC is appointed receiver [FDIC \(2023\)](#); USD Coin held part of its reserves there [Diop, Chevallier & Sanhaji \(2024\)](#)
- Weekend: redemptions could not settle, the band disappeared; USD Coin's low 0.8774 USD on Saturday 11 March (close 0.9715); Dai, crypto-backed, fell to 0.8970
- Tether rose to 1.0077 (high 1.0296): a flight to the other large stablecoin
- Sunday 12 March: all SVB depositors protected [Treasury, Federal Reserve & FDIC \(2023\)](#); the peg returned within two days: the event tested the backing, not the design

 [MFM_ch16_stablecoins](#)

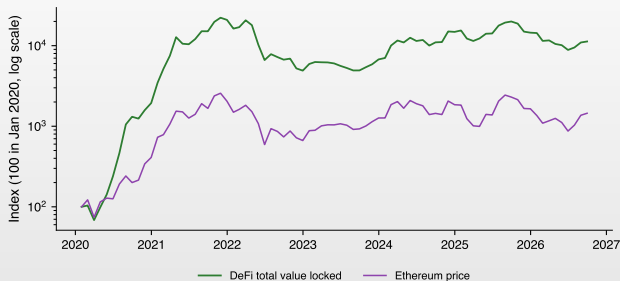
TerraUSD: an Algorithmic Stablecoin Collapses



- ▣ Supply and implied price (market value / supply), March–May 2022 ([DefiLlama](#))
- ▣ TerraUSD (UST) held its peg by exchange with the token LUNA, not with reserves; demand came from the high yields of the Anchor lending protocol [Liu, Makarov & Schoar \(2023\)](#)
- ▣ Supply 18.8 billion on 7 May 2022, 13.7 billion four days later (27% redeemed); implied price 0.754 USD on 10 May 2022
- ▣ Redemption minted more LUNA and pushed its price down: a death spiral that wiped out 50 billion USD in three days; sophisticated holders ran first and lost less [Liu, Makarov & Schoar \(2023\)](#)
- ▣ Without external collateral the peg rests on confidence alone: a run is self-fulfilling

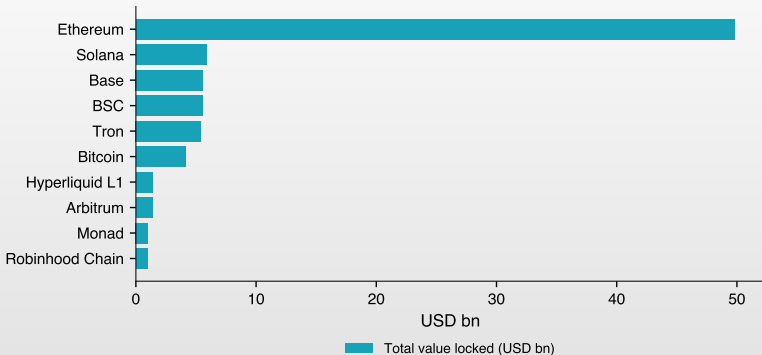
 [MFM_ch16_stablecoins](#)

Total Value Locked in DeFi



- Total value locked (TVL): the value of tokens deposited in DeFi protocols, all chains ([DefiLlama](#)), and the Ether price; month ends, 100 in January 2020, log scale
- TVL: 0.6 billion USD in January 2020, 177 billion at the peak (9 Nov 2021), 88 billion today (50% below the peak)
- Monthly $\Delta \ln \text{TVL}_t = a + b \Delta \ln P_t^{\text{ETH}} + u_t$, Newey–West SE: $\hat{b} = 0.79$ (SE 0.07), $R^2 = 0.62$, $n = 80$
- TVL mixes token prices with net deposits: measure activity with fixed-price quantities, net flows or fees; DeFi risks: leverage, liquidity mismatch, interconnection [Aramonte, Huang & Schrimpf \(2021\)](#)

Where the Value Is Locked



- TVL by blockchain, ten largest, 18 September 2026 ([DefiLlama](#)): Ethereum 49.8 billion USD, 57% of 88 billion; Solana 5.9, Base 5.6, BSC (BNB Smart Chain) 5.6, Tron 5.4
- Base and Arbitrum are layer-2 networks that settle on Ethereum: cheaper transactions, the same security anchor

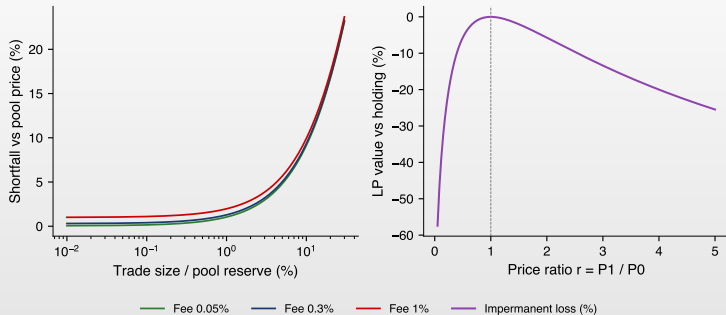
The Constant-Product Market Maker

- A pool holds reserves x of token X (e.g. Ether) and y of token Y (e.g. USD Coin); it accepts any trade that keeps $x \cdot y = k$ [Adams, Zinsmeister & Robinson \(2020\)](#)
 - ▶ Pool price (marginal): $p = y/x$ units of Y per X
- A trader sells Δx ; the pool keeps a fee f , so $\gamma \Delta x$ enters the invariant, $\gamma = 1 - f$
 - ▶ $(x + \gamma \Delta x)(y - \Delta y) = xy \Rightarrow \Delta y = \frac{y \gamma \Delta x}{x + \gamma \Delta x}$
- Execution price $\Delta y / \Delta x = p \frac{\gamma}{1 + \gamma \delta}$, with $\delta = \Delta x / x$ the trade size relative to the reserve
 - ▶ **Shortfall** against the pool price: $1 - \gamma / (1 + \gamma \delta)$: fee plus price impact
- Arbitrageurs keep p in line with other exchanges: the pool price follows the market [Angeris et al. \(2019\)](#)

Worked Example: a Swap in an Ether Pool

- Pool: $x = 2\,000$ Ether, $y = 5\,000\,000$ USD Coin, so $p = 2\,500$ and $k = 10\,000 \times 10^6$; fee $f = 0.3\%$
 - ▶ Sell $\Delta x = 20$ Ether ($\delta = 1\%$); after the fee, $\gamma \Delta x = 19.94$
- $\Delta y = \frac{5\,000\,000 \times 19.94}{2\,000 + 19.94} = 49\,358$ USD Coin, an execution price of 2467.9 per Ether
- Shortfall: $1 - 2467.9/2\,500 = 1.28\%$ (0.3% fee plus about 1% price impact)
- New pool price: $(5\,000\,000 - \Delta y)/2\,020 = 2450.8$, i.e. -1.97%
 - ▶ An arbitrageur now buys Ether in the pool and sells it elsewhere until p is back inside the no-arbitrage band that the fee sets around the market price (Seminar 16, A3)

Slippage and Impermanent Loss

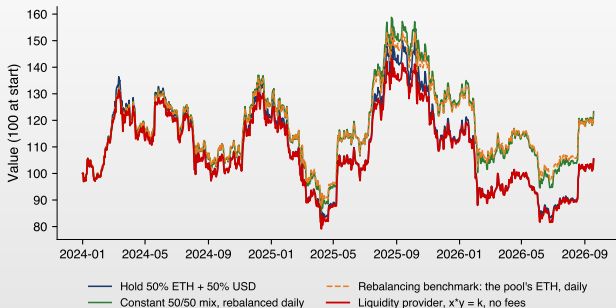


- Left: shortfall by trade size and fee; with a 0.3% fee 0.40% for 0.1% of the reserve, 1.28% for 1%, 9.34% for 10% (0.05% fee: 1.04% at 1%)
- Small trades pay the fee, large ones the price impact (about δ): liquidity is the depth of the reserve, not an order book ([Chapter 10](#)); concentrated liquidity deepens the pool near the price ([Uniswap v3](#))
- Right: value of a liquidity position relative to holding the two tokens (next slide)

Impermanent Loss and Loss Versus Rebalancing

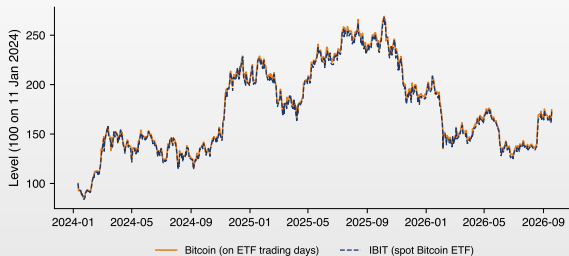
- A **liquidity provider** (LP) deposits x_0 and $y_0 = p_0 x_0$ (half and half in value) and receives a share of the pool
 - ▶ Without fees, arbitrage moves the pool to the market price p_1 : from $xy = k$ and $y/x = p_1$, $x_1 = \sqrt{k/p_1}$, $y_1 = \sqrt{k p_1}$
- LP value: $V_{LP} = p_1 x_1 + y_1 = 2\sqrt{k p_1}$; holding the tokens: $V_H = p_1 x_0 + y_0$
- With $r = p_1/p_0$ and $k = p_0 x_0^2$: **impermanent loss** $IL(r) = \frac{V_{LP}}{V_H} - 1 = \frac{2\sqrt{r}}{1+r} - 1 \leq 0$
 - ▶ $r = 2$ or $r = 0.5$: -5.72%; $r = 4$ or 0.25: -20.00%; symmetric in $\ln r$
- “Impermanent”: it vanishes if the price returns to p_0 ; it becomes permanent when the LP withdraws
- A fairer benchmark than holding: a **rebalancing** portfolio that holds the pool’s quantity of Ether at every instant and trades at market prices [Milionis et al. \(2022\)](#)
 - ▶ The pool rebalances too, but at stale prices set by arbitrageurs: it always trades at a loss to them
- For $x \cdot y = k$ and a price with volatility σ , the loss versus rebalancing (LVR) grows at the rate $LVR/V = \sigma^2/8$ per unit of time [Milionis et al. \(2022\)](#)
- Fees must exceed $\sigma^2/8$ a year for a liquidity provider to beat this benchmark (not to earn a positive return): a volatility-selling business
- Derivation by Itô in the [Appendix](#); a weighted pool $x^w y^{1-w} = k$ loses at the rate $\sigma^2 w(1-w)/2$ (Seminar 16, A6)

A Liquidity Provider in an Ether Pool



- Daily Ether prices from 1 January 2024; 100 invested half in Ether, half in USD; pool with no fees; dashed: the LVR benchmark (the pool's Ether, rebalanced daily)
- Ether 2352 $\rightarrow$ 2611 USD ($r = 1.11$): impermanent loss only -0.14%; final values: holding 105.5, pool 105.4, LVR benchmark 122.4
- Realised LVR 5.83% a year against $\sigma^2/8 = 5.79\%$ ($\sigma = 68.1\%$); 95% block-bootstrap CI [4.88, 6.98]%
- Impermanent loss vanishes when the price returns; LVR accumulates with every move: fees must earn about 5.79% a year to match the benchmark

IBIT Follows Bitcoin



- Bitcoin and IBIT on ETF trading days, both 100 on 11 Jan 2024
- A **spot ETF** holds the coins with a custodian and trades in US hours; BlackRock's IBIT charges a sponsor fee of 0.25% a year [BlackRock \(2026\)](#)
- **Authorised participants** create or redeem shares against coins, which keeps the price within a cost band around the net asset value, as for a stablecoin

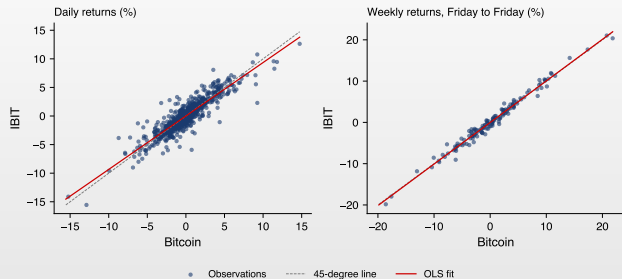
Measuring How Well an ETF Tracks

- Regression on the ETF's trading days: $r_t^{\text{ETF}} = a + \beta r_t^{\text{coin}} + u_t$; a perfect tracker has $\beta = 1$, $R^2 = 1$
- Tracking error** $\text{TE} = \text{sd}(r_t^{\text{ETF}} - r_t^{\text{coin}})\sqrt{m}$ ($m = 252$ daily, 52 weekly); **fee drift**: slope of $\ln(P_t^{\text{ETF}}/P_t^{\text{coin}})$ on time in years (the fund sells coins to pay its fee)

	$\hat{\beta}$ daily	R^2 daily	$\hat{\beta}$ weekly	R^2 weekly	TE daily	TE weekly
IBIT / Bitcoin	0.93	0.82	1.01	0.98	20.8	6.2
ETHA / Ether	0.94	0.83	0.99	0.98	29.9	9.0

- Weekly data show a near-perfect tracker; fee drift -0.24% a year (IBIT) and -0.24% (ETHA), close to the 0.25% fee
- IBIT trades 2.2 billion USD a day on average (maximum 10.3 billion on 5 Feb 2026)

Daily vs Weekly Returns and the Dimson Beta



▣ IBIT against Bitcoin: daily returns on common days (left), Friday-to-Friday returns (right); R^2 0.82 vs 0.98

▣ The coin's daily close and the New York close are hours apart: part of day t 's move reaches the ETF on $t \pm 1$, which biases the daily OLS beta towards zero; Newey–West fixes the SE, not the bias

▣ Dimson (1979):

$$r_t^{\text{ETF}} = a + \sum_{j=-1}^1 b_j r_{t+j}^{\text{coin}} + u_t, \text{ sum beta}$$

$$\hat{\beta}_D = \hat{b}_{-1} + \hat{b}_0 + \hat{b}_1$$

▣ IBIT: $\hat{\beta}_D = 1.00$ (SE 0.023) vs OLS 0.94; ETHA: 0.99 (SE 0.022) vs 0.94: within one SE of 1, as the weekly betas; where prices form is a separate question Hasbrouck (1995)

 MFM_ch16_etf

Crypto Equities: Coinbase and Strategy

- ▣ **Coinbase**: a US crypto exchange, revenue tied to trading volume; **Strategy** (formerly MicroStrategy): a software company holding Bitcoin as its main treasury asset

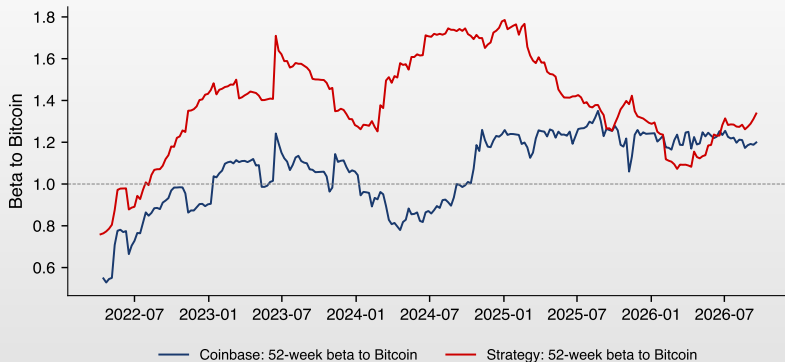
- ▣ Weekly two-factor regression, Newey–West SE, April 2021 – September 2026:

$$r_t^{\text{stock}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{S\&P}} + u_t$$

	$\hat{\beta}_{\text{BTC}}$	SE	$\hat{\beta}_{\text{SPX}}$	SE	R^2	σ_a (%)
Coinbase	0.79	0.09	2.05	0.22	0.52	84
Strategy	1.18	0.11	1.02	0.24	0.63	89

- ▣ Strategy: 1.18 times Bitcoin plus 1.02 times the S&P 500, a leveraged Bitcoin position with extra equity risk
- ▣ Coinbase: $\hat{\beta}_{\text{SPX}} = 2.05$, a high-beta US stock whose business depends on crypto activity, more an equity than a Bitcoin proxy

Beta to Bitcoin Over Time



- Rolling 52-week beta of weekly returns on Bitcoin; Strategy's ranged from 0.76 to 1.79; both stocks are more volatile than Bitcoin (55% a year on the same weeks)
- When did the beta change? A break test at an unknown date: Seminar 16, B7–B8

Testing a Factor Model: GRS and Fama–MacBeth

- N test assets, K traded factors, T periods: $r_{it}^e = \alpha_i + \beta_i' f_t + \varepsilon_{it}$; model adequacy $H_0: \alpha_1 = \dots = \alpha_N = 0$
 - ▶ $GRS = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F(N, T - N - K)$ [Gibbons, Ross & Shanken \(1989\)](#); $\hat{\Sigma}$, $\hat{\Omega}$: residual and factor covariances
 - ▶ F is exact only for i.i.d. errors with the Normal distribution: with crypto tails, add a wild bootstrap under H_0
- Premium of a factor: Fama–MacBeth, two passes; $\hat{\lambda}$ = mean of the weekly cross-sectional slopes on $\hat{\beta}_i$
 - ▶ The FM standard error ignores that $\hat{\beta}_i$ is estimated: [Shanken \(1992\)](#) correction; weak factors (small spread of β) over-reject [Kleibergen \(2009\)](#)
- GRS judges the model, FM the premium: a priced factor in an incomplete model can fail GRS; formulas and robust versions in the [Appendix](#)

Worked Result: Eight Coins and the Crypto Market Factor

- Test assets: Bitcoin, Ethereum, XRP, BNB, Cardano, Dogecoin, Litecoin, Chainlink; weekly excess returns over the 4-week bill ([FRED DTB4WK](#)), $T = 450$; factor: the five-coin index of Section 3
 - ▶ CSMB and CMOM of [Liu, Tsyvinski & Wu \(2022\)](#) need long–short portfolios over a broad cross-section with circulating market values: not buildable here
- GRS = 2.16: $p = 0.029$ from $F(8, 450 - 9)$, $p = 0.018$ from the wild bootstrap: zero alphas rejected at 5%
 - ▶ Largest alphas: Dogecoin +99% a year ($t = 1.23$), BNB +51% ($t = 1.97$); no single $|t|$ above 1.97
- Fama–MacBeth: $\hat{\lambda} = 163\%$ a year, SE 147 (FM) and 156 (Shanken); the factor itself earned 39% ($t = 1.81$)
 - ▶ Betas range only from 0.90 to 1.40: eight coins cannot identify a premium
- Survivorship (coins chosen because they are large today) may contribute to the rejection; GRS does not identify its cause

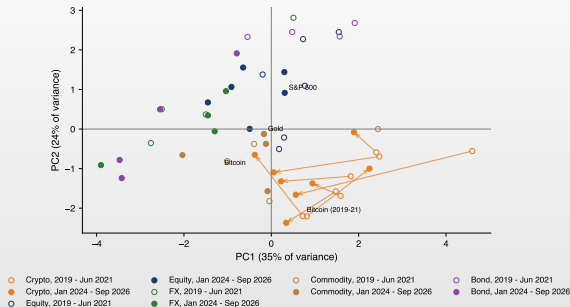
A Separate Asset Class? The Evidence So Far

- [Pele et al. \(2023\)](#) compare cryptos with stocks, exchange rates, commodities, bonds and real-estate indices
 - ▶ Dimension reduction: most of the variation is explained by **tail**, **memory** and **moment** factors of log returns
 - ▶ Classification methods place cryptos in a separate class, mainly because of the tail factor
 - ▶ The tail factor is a feature-space factor (76% of the variance), loading on quantiles, conditional tail expectations, variance and the α -stable tail and scale parameters ([Pele et al. \(2023\)](#), Sect. 3.2): it mixes scale with shape and is not a difference in Hill indices
 - ▶ Over time cryptos became more alike and more distinct from other assets
- Hedge or safe haven? A hedge is uncorrelated or negatively correlated on average; a safe haven on the worst market days [Baur & Lucey \(2010\)](#)
- Early evidence: Bitcoin a diversifier, a hedge only in some periods [Bouri et al. \(2017\)](#); tested with post-ETF data in Seminar 16, C1

A Small Replication: Features and Method

- ▣ 26 assets: 8 cryptos, 6 equity indices (including BET), 4 exchange rates, 4 commodities, 4 bond ETFs; two windows
 - ▶ W1: January 2019 – June 2021; W2: 11 January 2024 – 18 September 2026 (after the spot ETFs)
- ▣ Eight features of daily log returns, each asset on its own calendar
 - ▶ Moments: $\ln \sigma_a$, skewness, $\ln(1 + K)$; memory: $\hat{\rho}_1(r)$, $\hat{\rho}_1(r^2)$
 - ▶ Tails: Hill indices of losses and gains (5% largest); maximum drawdown
- ▣ **Principal component analysis** (PCA) of the standardised features ([Chapter 3](#)): directions of maximum variance
 - ▶ Distance between class centroids in the full 8-dimensional standardised space

Cryptos Among Other Asset Classes



- First two principal components (35% and 24% of the variance); hollow: W1 (2019 – June 2021), filled: W2 (January 2024 – September 2026); arrows: each crypto
- PC1 measures volatility, kurtosis, drawdown depth and tail heaviness; cryptos sit apart in both windows, as in [Pele et al. \(2023\)](#)
- Point estimates move towards the others: distance to equities $3.03 \rightarrow 2.63$, to commodities $2.26 \rightarrow 1.86$; Bitcoin's volatility $77\% \rightarrow 47\%$
- Cryptos became more alike (spread around their centre $2.13 \rightarrow 1.43$); median Hill index $2.32 \rightarrow 3.26$: larger than sampling noise? Next slide

 MFM_ch16_alt_assets

Is the Move Significant? A Block Bootstrap of the Map

- In each window, draw the same calendar days for all 26 assets (moving blocks of 28 days: four weeks, 20 trading days); each asset keeps its own returns on those days, so cross-asset dependence is preserved; recompute the eight features, the common standardisation and the centroid distances; 500 replications
 - ▶ Statistic: change $\Delta = d_{W2} - d_{W1}$ in the distance between the crypto centroid and each class centroid
- Equities: $\Delta = -0.39$, 95% CI $[-1.04, 0.59]$; commodities: -0.41 $[-1.38, 0.47]$; both contain 0
 - ▶ Currencies: -0.89 $[-1.67, 0.18]$; bonds: -0.62 $[-1.55, 0.92]$
 - ▶ Spread of the cryptos around their centre: -0.70 $[-1.09, 0.31]$
- Reading: all five indicative intervals contain 0; the map shows cryptos apart, but it does not establish a reliable post-ETF move towards the other classes
- Caveat: $\ln(1 + K)$ and $\hat{\rho}_1(r^2)$ have population targets only if $Er^4 < \infty$, doubtful here: they enter as sample descriptors, so the map is descriptive and these intervals are indicative

Case Study: Three Factors for Crypto Returns

- Liu, Tsyvinski & Wu (2022), *Journal of Finance*: a landmark study that applies the equity factor-model toolkit (portfolio sorts, factor regressions, GRS) to the whole cross-section of coins
 - ▶ Question: do a few crypto-specific factors price the expected returns of all coins, as market, size and momentum do for stocks?
- It builds on Liu & Tsyvinski (2021): aggregate crypto returns barely load on stock, currency or metal factors, but are predicted by crypto momentum and attention
 - ▶ The step: from the time series of three coins to the cross-section of 1,827 coins
- Why it matters: a factor model is the benchmark for every claimed crypto anomaly and for judging crypto funds
 - ▶ If coins are segmented from equities, stock factors are the wrong benchmark
- Link to this chapter: the market-value index of Section 3; CMKT, CSMB, CMOM, the GRS test of Section 7; data-quality threats in the [Appendix](#)

Case Study: Data and Method

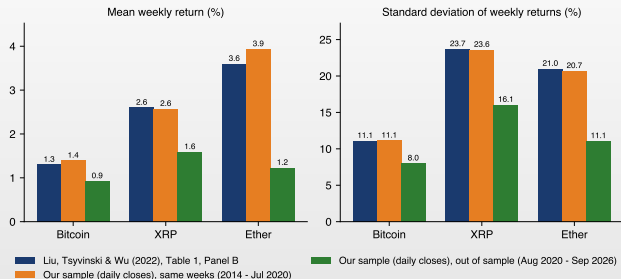
- Data: daily prices, volumes and market values of active and defunct coins from an aggregator of over 200 exchanges (Liu, Tsyvinski & Wu (2022), Sect. 2); market value of at least 1 million USD: 1,827 coins, first week of 2014 – July 2020
 - ▶ Weekly returns from daily closes: 52 weeks a year, the last one of eight days (nine in leap years); CMKT: value-weighted return of all coins minus the one-month bill rate
- Anomalies: 24 characteristics from the equity literature that need only price, volume and market value (Table 2); each week, value-weighted quintile portfolios and a top-minus-bottom spread
 - ▶ CSMB: bottom 30% minus top 30% of coins by market value; $CMOM = \frac{1}{2}(SH + BH) - \frac{1}{2}(SL + BL)$ from 2×3 sorts on size and past three-week return (S/B: small/big coins; H/L: high/low past three-week return; Liu, Tsyvinski & Wu (2022), Sect. 4.1, Eq. 1)
- Tests: the ten significant spreads regressed on one-, two- and three-factor models (Tables 8–9); joint GRS tests on all 24 spreads (Table 11)
 - ▶ Multiple testing: p -values adjusted under a generalised FWER (k -FWER, probability of k or more false rejections, Lehmann & Romano (2005)), weaker than Holm's FWER

Case Study: Key Results

Result	Numbers in the paper	Reading
Weekly returns (Table 1)	mean, s.d.: market 1.3%, 11.2%; Bitcoin 1.3%, 11.1%; XRP 2.6%, 23.7%; Ether 3.6%, 21.0%	the market behaves like Bitcoin
Size (Table 3)	top minus bottom quintile -5.8% a week ($t = -2.45$)	premium in the smallest coins
Momentum (Table 4)	three-week winners minus losers 3.1% a week ($t = 2.65$)	winners keep winning
Crypto CAPM (Table 8)	all ten alphas significant; momentum $\alpha = 3.0\%$ ($t = 2.55$), $R^2 = 0.001$	the market alone fails
Three factors (Table 9)	no alpha significant; momentum $\alpha = 1.1\%$ ($t = 1.41$), CMOM loading 1.12, $R^2 = 0.58$	size and momentum explain the ten anomalies
Joint test (Table 11)	GRS on 24 spreads: $p = 0.033$ (Fama–French six factors), $p = 0.194$ (crypto three)	stock factors are the wrong benchmark

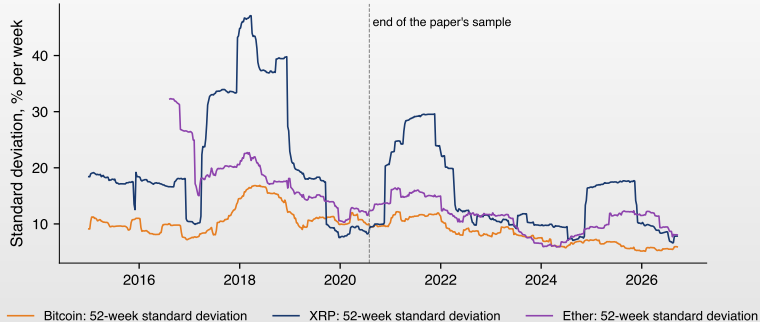
- Momentum is a large-coin effect: 3.2% a week above the median size, insignificant below (Table IA.10), the opposite of stocks

Case Study: Table 1 Rebuilt on the Course Data



- Weekly simple returns on the paper's calendar; blue: Table 1, Panel B of [Liu, Tsyvinski & Wu \(2022\)](#); orange: our sample (daily closes), same weeks; green: August 2020 – 16 Sep 2026
- Same weeks: mean Bitcoin 1.4% (paper 1.3%), XRP 2.6% (2.6%), Ether 3.9% (3.6%); s.d. 11.1, 23.6, 20.7% (paper 11.1, 23.7, 21.0)
- Largest gap, the Ether mean: 0.26 i.i.d. standard errors (SE 1.29 points); this checks the data and the calendar, not the factors
- Out of sample (319 weeks): means 0.9, 1.6, 1.2%; s.d. 8.0, 16.1, 11.1%: Ether's volatility roughly halves

Case Study: Volatility Before and After the Paper's Sample



- Standard deviation of weekly returns over the last 52 weeks; dashed: end of July 2020, the end of the paper's sample; peaks: XRP 47.1% (Mar 2018), Bitcoin 16.9% (May 2018) a week
- Last window: Bitcoin 5.9%, XRP 7.8%, Ether 8.0% a week; volatility moves in regimes: factor tests across 2020 need HAC errors and a break test at an unknown date (Seminar 16, C3)

Case Study: Replicate, Limits, Extension

- **Replicate** with the course data: daily closes of the nine non-stable coins (Bitcoin, Ether, XRP, BNB, Cardano, Solana, Dogecoin, Litecoin, Chainlink); one-month Treasury bill rate from the [Kenneth French Data Library](#)
 - ▶ Table 1, Panel B for Bitcoin, XRP and Ether: previous slides; next, the median, skewness and kurtosis, and the CMKT proxy: the five-coin market-value index of Section 3
 - ▶ Data limit: CSMB and CMOM need breakpoints over hundreds of coins with market values, dead coins included; the nearest published design is the top-20 tercile of Table IA.13, but our nine coins are today's survivors
- **Limits** stated by the authors (Section 6.2)
 - ▶ A short sample, 2014 – July 2020, while investors were still learning about the technology: only the split-half check (same signs)
 - ▶ Most coins cannot be shorted; for the top 20 coins, after trading costs, about 60% of the returns of long-short strategies that short Bitcoin remain (Table IA.16)
 - ▶ With prices from three reputable USD exchanges, the market-value size spread is no longer significant (Table IA.15)
- **Open extension:** out of sample, August 2020 – September 2026 (the 2022 collapses, the spot ETFs), does three-week momentum among large coins survive, and is CMKT still segmented from equities?
 - ▶ Crypto-CAPM alphas; the segmentation regression of footnote 16 of [Liu, Tsyvinski & Wu \(2022\)](#) on the global five factors [Fama & French \(2015\)](#) and five exchange rates; a break test at an unknown date (Seminar 16, C3)

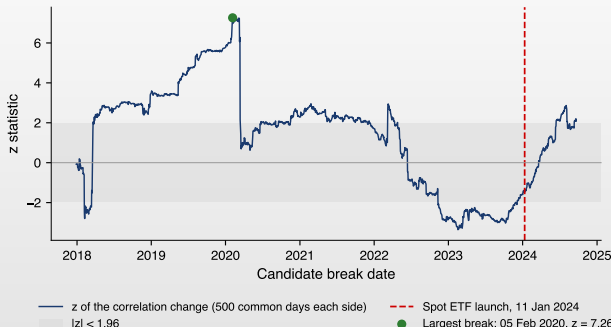
An Open Question and the Discovery Loop

- Did the US spot Bitcoin ETFs (since 11 January 2024) integrate Bitcoin with equities, in the mean, in the tails and in price discovery?
 - ▶ A live question: [Lee & Lee \(2025\)](#) study the shifted links after the approval; [Kang & Lee \(2026\)](#) title their answer “institutionalization without integration”
- Formal version: $H_0: \rho_{\text{post}} = \rho_{\text{pre}}$ against $H_1: \rho_{\text{post}} > \rho_{\text{pre}}$; the same for the lower-tail dependence λ_L of [Chapter 6](#)
 - ▶ Falsified if the data choose a break far from January 2024, or if placebo dates give changes as large
- Identification: the launch is an event, not an experiment; it overlaps with the rate cycle, the April 2024 halving and flows that respond to prices
- The discovery loop in one pass, the assistant drafts and the researcher verifies
 - ▶ literature (resolve every DOI), competing hypotheses, code that first replicates $\hat{\rho} = 0.39$ and 0.38, a pre-registered robustness plan, a referee-style critique; reusable prompts in the [Appendix](#)

What the Researcher Must Check

- ▣ **Calendars:** returns computed on each calendar and then matched pair Bitcoin's Sunday-to-Monday move with the S&P 500's Friday-to-Monday move; align prices on common trading days first, then compute returns
- ▣ **Closing times and volatility:** non-synchronous closes bias daily correlations towards zero, high volatility raises them [Forbes & Rigobon \(2002\)](#): check weekly returns, the lead-lag correlation and the adjusted ρ^*
- ▣ **Survivorship and supply:** use the universe as it was on each date and the circulating supply, as in the CRIX rule
- ▣ **Invented references and numbers:** every DOI resolves on Crossref with the same title; every number is regenerated by code in the repository

Mini-Case: Did the ETFs Break the Correlation?



- ▣ Assistant's claim: "the spot ETFs integrated Bitcoin with equities"; $\hat{\rho} = 0.39$ in 2020–2023 ($n = 1006$), 0.38 since 11 January 2024 ($n = 674$)
- ▣ Difference -0.01 , 95% CI $[-0.12, 0.13]$ (circular block bootstrap, blocks of 20 days)
- ▣ Chart: $z(\tau) = \frac{[\text{atanh } \hat{\rho}_{\text{after}} - \text{atanh } \hat{\rho}_{\text{before}}]}{\sqrt{2/(h-3)}}$, $h = 500$ days on each side
- ▣ Largest break 5 Feb 2020 ($z = 7.26$); at the ETF date $z = -1.48$ and 82% of candidate break dates give a larger $|z|$: the integration came in early 2020; tails and price discovery remain open

 ai_discovery_case.py

Your Turn: a Project Seed

- ▣ **Question:** did institutionalisation change Bitcoin's **tail** dependence and **price discovery** with US equities?
- ▣ **Replicate first:** $\hat{\rho} = 0.39$ and 0.38 from this chapter, then one headline estimate of [Lee & Lee \(2025\)](#) or [Kang & Lee \(2026\)](#) on their own sample
 - ▶ Extend: lower-tail dependence from copulas ([Chapter 6](#)), weekly returns, IBIT–Bitcoin lead–lag in US trading hours
- ▣ **Deliverables:** a pre-registered plan committed before the results, a notebook that regenerates every number, effect sizes with CIs and the full list of specifications
- ▣ **AI log:** prompts that shaped the design in `AI_USE.md`; each wrong claim (an invented DOI, returns on mismatched calendars) and the test that caught it in `AI_ERRORS.md` ([templates](#))

Key Takeaways

- Crypto returns: 3–7 times the volatility of shares, drawdowns of 80–96%, heavy tails in all six assets (Hill near 2.7), calmer weekends; the link with equities broke in March 2020, not at the ETF launch
- Indices: a Laspeyres-type index with a divisor; CRIX picks its size by an AIC on tracking residuals; value weights make crypto indices mostly Bitcoin
- Stablecoins: the peg is an arbitrage band that holds only while redemption works; a threshold AR fits it, tested with a bootstrap, not with χ^2
- DeFi: $xy = k$ prices a swap as fee plus price impact; providing liquidity is selling volatility, at $\sigma^2/8$ a year
- ETFs track one for one once closing times are handled; cryptos remain a separate class in distribution, a risk asset in co-movement

Check Yourself, and Next: Chapter 17

Retrieval questions

- Why is Bitcoin's sample kurtosis not reported, and what replaces it?
- Why can a threshold or a break date not be tested with χ^2 critical values?
- Why does a daily regression understate how well IBIT tracks Bitcoin?

Next chapters

- [Chapter 17](#), bubbles and crashes: explosive-root tests and log-periodic power laws, crypto booms as a main case
- [Chapter 18](#), systemic risk: spillovers between crypto, banks and equity markets
- Revisit: VaR and ES for crypto ([Chapter 7](#), [Pele & Mazurencu-Marinescu-Pele \(2019b\)](#)), liquidity on crypto exchanges ([Chapter 10](#), [Agakishiev, Härdle & Pele \(2025\)](#)); reading: [Härdle, Harvey & Reule \(2020\)](#), [Franke, Härdle & Hafner \(2019\)](#), [Schär \(2021\)](#)

Appendix: Tail Index, Inference Conditions

- Under i.i.d. data with a regularly varying tail, $k \rightarrow \infty$, $k/n \rightarrow 0$, and a second-order condition with negligible bias ($\sqrt{k} A(n/k) \rightarrow 0$): $\sqrt{k}(\hat{\alpha}_H - \alpha) \Rightarrow N(0, \alpha^2)$; the 5% share is a finite-sample tuning choice
 - ▶ With volatility clustering the variance depends on extremal dependence and is typically larger [Drees \(2000\)](#); a block bootstrap needs further tail and block-length conditions; choose k from a Hill plot [Drees, de Haan & Resnick \(2000\)](#)
 - ▶ Bitcoin: $\hat{\alpha}_H = 2.73$ ($k = 78$), SE 0.31; with $k = 2.5\%$ or 10% : 3.48 and 2.59: the choice of k matters as much as the sampling error
- Block bootstrap intervals: up to 4.48 for the S&P 500; bootstrap CIs for the sample kurtosis are invalid when $\alpha < 4$ (bootstrap failure with infinite variance, [Athreya \(1987\)](#))
 - ▶ Dogecoin: sample kurtosis 106.4, driven by a few social-media jumps, with $\hat{\alpha}_H = 2.83$: it estimates nothing stable
- Equal tails? Bitcoin vs S&P 500: $z = -0.28$; Dogecoin vs S&P 500: $z = -0.07$ (i.i.d. SE, independent estimates)
 - ▶ Overlapping CIs are not a test of equality, and non-rejection is not equivalence: that needs a dependence-aware interval for the difference, robust to the choice of k

Appendix: The Published CRIX Rule in Detail

- Total market index TMI (all coins) and candidate indices $\text{CRIX}(k, \beta)$: the k_1 largest coins at weight 1, the next $s = k - k_1$ with free weights β (Trimborn & Härdle (2018), Sect. 3, eqs. (11)–(12) in Trimborn & Härdle (2018, author version))
 - ▶ $\hat{\varepsilon}_t(k, \beta) = \Delta \ln \text{TMI}_t - \Delta \ln \text{CRIX}_t(k, \beta)$; $\hat{\beta}$ minimises $\|\hat{\varepsilon}(k, \beta)\|^2$
- $\text{AIC}(k) = -2 \log L\{\hat{\varepsilon}(k, \hat{\beta})\} + 2s$ (eq. (20)); L uses an Epanechnikov kernel density of $\hat{\varepsilon}(k_1, 1)$ with a Sheather–Jones plug-in bandwidth
 - ▶ Under Normal errors it reduces to $T \ln \hat{\sigma}_k^2 + 2s$ (Trimborn & Härdle (2018, author version), Appendix, eq. (35))
- ECRIX: $k_1 = 1$, steps of one coin, stop at the first rise of the AIC (Trimborn & Härdle (2018, author version), Sect. 5, rule (26)); EFCRIX: global minimum (rule (27))
 - ▶ Members reallocated monthly by market value; k reviewed quarterly on the last three months of data (Trimborn & Härdle (2018), Sect. 4)
- Data limitation: circulating supply is public for five of our coins only (Bitcoin, Ethereum, Dogecoin, Cardano, Litecoin), so $k \leq 4$ and TMI has five members, not hundreds

Appendix: Peg Deviations, Summary Statistics

	s.d. (bp)	$ d_t > 50$ (%)	lowest close (bp)	lowest low (bp)
Tether	7.3	0.14	-41	-515
USD Coin	7.5	0.14	-285	-1226
Dai	10.8	0.57	-261	-1030

□ Daily close, 1 January 2021 – 18 September 2026; $d_t = 10^4(P_t - 1)$ in basis points; lowest low from the daily low price

Appendix: Tokenisation of Real-World Assets

- ▣ **Tokenised yield-bearing instruments:** claims on Treasury bills and cash issued as tokens on a blockchain; the legal form differs
 - ▶ Fund shares: BlackRock BUIDL 2.38, Circle USYC 2.55; a tokenised secured note issued by a special-purpose vehicle: Ondo USDY 2.22 ([Ondo](#)) (billion USD, 18 Sep 2026, [DefiLlama](#))
- ▣ Unlike a payment stablecoin, they pay the Treasury yield to the holder
- ▣ Benefits: settlement 24/7, use as collateral in DeFi; risks: smart-contract bugs, legal ownership, liquidity in stress [Aramonte, Huang & Schrimpf \(2021\)](#)
- ▣ Token economics and adoption: [Cong, Li & Wang \(2021\)](#)

Appendix: Decentralised Finance in Five Layers

- Schär (2021) describes DeFi as a stack of layers
 - ▶ **Settlement:** the blockchain and its native coin (Ether)
 - ▶ **Assets:** tokens issued on it (stablecoins, governance tokens)
 - ▶ **Protocols:** exchanges, lending, derivatives as smart contracts
 - ▶ **Applications and aggregators:** user interfaces that route trades across protocols
- Promise: open access, transparency, composability (protocols plug into each other)
- Risks: high leverage, liquidity mismatches, built-in interconnectedness, no shock absorber [Aramonte, Huang & Schrimpf \(2021\)](#)
 - ▶ “Decentralisation illusion”: governance tends to concentrate power [Aramonte, Huang & Schrimpf \(2021\)](#)

Appendix: Deriving $LVR = \sigma^2/8$

- Pool value at market price P after arbitrage: $V(P) = Px + y = 2\sqrt{kP}$, concave: $V'(P) = \sqrt{k/P} = x$, $V''(P) = -\frac{1}{2}\sqrt{k}P^{-3/2}$
 - ▶ Rebalancing portfolio: holds the same $x_t = V'(P_t)$ Ether at every instant, so $dR_t = V'(P_t) dP_t$
- Price $dP_t = \mu P_t dt + \sigma P_t dW_t$; Itô: $dV_t = V' dP_t + \frac{1}{2} V'' \sigma^2 P_t^2 dt$
 - ▶ $d(R_t - V_t) = -\frac{1}{2} V''(P_t) \sigma^2 P_t^2 dt = \frac{1}{4} \sqrt{kP_t} \sigma^2 dt = \frac{\sigma^2}{8} V_t dt$
- LVR is the pool's negative gamma: a short-volatility position that loses $\sigma^2/8$ of its value per unit of time, whatever the drift [Milionis et al. \(2022\)](#)
- Assumptions: continuous geometric Brownian motion, frictionless arbitrage, no fees, continuous blocks
 - ▶ With fees and discrete blocks, arbitrageurs trade only when the price leaves the fee band, which lowers the loss [Milionis, Moallemi & Roughgarden \(2025\)](#)
 - ▶ Weighted pool $x^w y^{1-w} = k$: rate $\sigma^2 w(1-w)/2$ (Seminar 16, A6)

Appendix: The Liquidity-Provider Benchmarks in Detail

- ▣ Realised LVR rate $\sum_t (\Delta R_t - \Delta V_t) / V_{t-1}$ per year; cumulative LVR 17.0 points over 2.7 years
- ▣ The match with theory is not a test: each day's loss is $R_t/2 - (\sqrt{1 + R_t} - 1) \approx R_t^2/8$, so the realised LVR is the realised variance over 8 (bootstrap correlation of the two: 0.9981)
- ▣ 2.7 years pin the realised rate down only to about ± 1 point: CI [4.88, 6.98]%
- ▣ The constant 50/50 mix (final value 123.3) is a different benchmark with the same instantaneous rate: its log growth exceeds the pool's by $\sigma^2/8$, here 5.79% a year

Appendix: Crypto Factors, the Literature

- [Liu & Tsyvinski \(2021\)](#): crypto returns have little exposure to stock, currency or commodity factors; they load on crypto momentum and investor attention
 - ▶ [Liu, Tsyvinski & Wu \(2022\)](#): three factors from weekly coin returns: CMKT (value-weighted market excess return), CSMB (small minus big coins), CMOM (past three-week winners minus losers)
- Value from on-chain adoption and segmentation from traditional markets [Cong, Karolyi, Tang & Zhao \(2026\)](#); mispricing versus risk compensation [Babiak & Bianchi \(2025\)](#)
- Fundamentals: network value follows Metcalfe's law in the long run, not in the short run [Pele & Mazurencu-Marinescu-Pele \(2019a\)](#); adoption and token value reinforce each other [Cong, Li & Wang \(2021\)](#)
- Two different questions, tested separately on the next slides
 - ▶ Model adequacy: are all test-asset alphas zero for the chosen factors? (time series, GRS)
 - ▶ Risk premium: is a factor's premium nonzero? (a traded factor: its mean excess return; cross-section: Fama–MacBeth λ)

Appendix: Time-Series Test, GRS

- N test assets, K traded factors, T periods: $r_{it}^e = \alpha_i + \beta_i' f_t + \varepsilon_{it}$; $H_0: \alpha_1 = \dots = \alpha_N = 0$
 - ▶ $GRS = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F(N, T - N - K)$ [Gibbons, Ross & Shanken \(1989\)](#)
 - ▶ $\hat{\Sigma}$: residual covariance, $\hat{\Omega}$: factor covariance (both divided by T); the denominator is one plus the squared Sharpe ratio of the factors
- The F distribution is exact only if the errors, conditional on the factors, are i.i.d. with the Normal distribution; crypto returns have $\hat{\alpha}_H$ near 3 and volatility clustering, so these assumptions are doubtful for the residuals
 - ▶ Robust alternatives: a GMM Wald test with HAC covariance, or a wild bootstrap under H_0 : $r_t^{e*} = \hat{\beta}' f_t + \hat{\varepsilon}_t \eta_t$, one Rademacher η_t for all assets (keeps the cross-correlation); valid if the residuals are serially uncorrelated (martingale differences), otherwise a block or dependent wild bootstrap
- $\hat{\Sigma}$ must be invertible and precise: N much smaller than T ; weekly crypto data since 2018 give $T \approx 450$
- GRS tests the joint pricing restriction, not a factor's premium: a zero-premium factor need not cause a rejection, and a priced factor in an incomplete model can fail GRS

Appendix: Cross-Section, Fama–MacBeth, Shanken and Weak Factors

- Two passes: $\hat{\beta}_i$ from time series; then each week $r_{it}^e = \gamma_0 + \hat{\beta}_i' \lambda_t + e_{it}$; $\hat{\lambda} = \bar{\lambda}_t$, $SE_{FM} = sd(\hat{\lambda}_t)/\sqrt{T}$
 - ▶ The FM standard error ignores that $\hat{\beta}_i$ is estimated (errors in variables): too small
 - ▶ [Shanken \(1992\)](#), for $\hat{\theta} = (\hat{\gamma}_0, \hat{\lambda}')'$ and $X = [1, \hat{\beta}]$: $\widehat{\text{Var}}(\hat{\theta}) = \frac{1}{T} [(X'X)^{-1}X'\hat{\Sigma}X(X'X)^{-1}(1 + \hat{\lambda}'\hat{\Sigma}_f^{-1}\hat{\lambda}) + \hat{\Sigma}_f^*]$,
 $\hat{\Sigma}_f^* = \begin{pmatrix} 0 & 0 \\ 0 & \hat{\Sigma}_f \end{pmatrix}$; $\widehat{\text{Var}}(\hat{\lambda})$ is its lower $K \times K$ block
 - ▶ Assumptions: residuals serially uncorrelated and homoskedastic conditional on the factors, $\hat{\Sigma}$ their covariance; the residual part is multiplied by $1 + \hat{\lambda}'\hat{\Sigma}_f^{-1}\hat{\lambda}$, the factor part $\hat{\Sigma}_f^*$ is added once
- Weak or useless factors (small spread of β , e.g. attention proxies): λ is weakly identified and FM t -statistics over-reject
 - ▶ Use identification-robust tests of risk premia [Kleibergen \(2009\)](#)
- Multiple testing: dozens of crypto “anomalies” have been tried; a new factor should clear $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#), or control the false discovery rate

Appendix: Data-Quality Threats to Identification

- **Wash trading:** fake volume on unregulated exchanges inflates volume- and liquidity-based factors [Cong, Li, Tang & Yang \(2023\)](#)
 - ▶ Filter the exchanges and volumes using the rules of the cited paper before building any factor
- **Which price?** Large, recurrent gaps between exchanges, much larger across countries than within them [Makarov & Schoar \(2020\)](#)
- **Survivorship:** today's large coins are the winners; dead coins are missing from most samples [Będowska-Sójka, Wójcik & Pele \(2026\)](#)
- **Remedy:** fix the investable universe ex ante (exchange list, volume filter), keep delisted coins, report results by sub-universe

Appendix: Common Pitfalls

- Reporting sample kurtosis as a population estimate when $\hat{\alpha}_H < 4$: if $\alpha < 4$ it estimates nothing and its bootstrap CI is invalid; assess first whether $Er^4 < \infty$ is plausible
- Testing a threshold or a break date with χ^2 critical values: the nuisance parameter is not identified under H_0
- Comparing daily closes recorded at different times and concluding that an ETF tracks badly
- Reading TVL growth as adoption without separating the valuation of the tokens from net deposits
- Treating a stablecoin as risk-free cash: the peg is only as good as redemption and reserves
- Total instead of circulating supply in a market-value index; survivorship: studying only today's large coins ignores the many that failed [Będowska-Sójka, Wójcik & Pele \(2026\)](#)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ `“List 2023-2026 papers that test for a change in Bitcoin-equity co-movement at the US spot ETF approval: DOI, sample, method, sign of the effect.”`
 - ▶ Resolve each DOI on [Crossref](#), open the paper, find the number it is cited for; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing mechanisms, each with an implication the others do not share
 - ▶ `“Give three competing mechanisms for the post-2020 Bitcoin-S&P 500 correlation and, for each, one testable implication that separates it from the others.”`
 - ▶ E.g. ETF flows predict an effect concentrated in US trading hours; a common monetary factor predicts co-movement with rate surprises already before 2024
- **Data and code:** on the course data, replicating a known number first
 - ▶ `“Take the daily closes of Bitcoin and the S&P 500 from the course data, align the prices on common trading days, then compute log returns and their 2020-2023 correlation.”`
 - ▶ Replicate first: this chapter reports 0.39 (2020–2023) and 0.38 (since the ETFs); no new result before the code hits both

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the post-ETF results are seen
 - ▶ “Draft a pre-registered robustness plan: break-date scan, placebo launch dates, weekly returns, tail dependence, block bootstrap; fix every tuning choice now.”
 - ▶ The plan fixes windows, block length and the family of tests: the number of specifications is known and the FWER can be controlled
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: give the three strongest reasons this change in correlation could be spurious, and the test that would settle each.”
 - ▶ Expect: non-synchronous closes, correlation that rises mechanically with volatility Forbes & Rigobon (2002), confounding events
- **Report**: effect size with a CI, every specification tried, what failed
 - ▶ The prompts that shaped the design go into AI_USE.md

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Introduction to Blockchain and Cryptocurrencies
- ▣  Comparison of Cryptocurrency Indices
- ▣  Cryptocurrency as an Asset Class
- ▣  Smart Contracts
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  Risk Premia in the Bitcoin Market

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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